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Paper:

Optimal Clustering of Point Cloud by 2D-LiDAR Using Kalman Filter

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Light detection and ranging (LiDAR) has been the primary sensor for autonomous mobility and navigation system owing to its stability. Although multiple-channel LiDAR (3D-LiDAR) can obtain dense point clouds that provide optimal performance for several tasks, the application scope is limited by its high-cost. When employing single channel LiDAR (2D-LiDAR) as a low-cost alternative, the quantity and quality of the point cloud cause conventional methods to perform poorly in clustering and tracking tasks. In particular, when handling multiple pedestrian scenarios, the point cloud is not distinguished and clustering is unable to succeed. Hence, we propose an optimized clustering method combined with a Kalman filter (KF) for simultaneous clustering and tracking applicable to 2D-LiDAR.

Keywords: 2D-LiDAR, autonomous mobility, Kalman filter, clustering, navigation

1. Introduction

Owing to the advancements in light detection and ranging (LiDAR), it can be adopted as an external sensor for tasks such as obstacle detection, navigation, and SLAM, which can be simply installed and utilized to develop an autonomous mobility platform. LiDAR with multiple channels (3D-LiDAR) is often a standard sensor because it can obtain dense point-cloud data, which is easy for post-processing (clustering, tracking, etc.); however, it is expensive. Even in the entry-level resolution, 3D-LiDAR (16 channels) remains overpriced and slow in promoting the process for an actual mobile society.

Accordingly, an optimal clustering and tracking method for single channel LiDAR (2D-LiDAR) in an automated mobility scooter was proposed. An autonomous mobility scooter is driven by a retrofitted set box based on our pre-work [1], which includes actuators and sensors, and then simply installed to make it automatic. The cost reduction of sensors is required to achieve low-cost automated driving. In addition, the field of testing (FOT) in our work was conducted at a shopping mall [2]. Although 3D-LiDAR is popular with small platforms because it can achieve a three-dimensional range sensing of the scenario [3], it is limited by its high computational cost. Hence, 2D-LiDAR was considered as an alternative. However, compared to 3D-LiDAR, 2D-LiDAR has few limitations, such as the sparse point cloud, which complicates clustering and tracking.

Moreover, the mobility scooters are considered pedestrians according to law and can be used indoors and outdoors. Hence, it is assumed that the elderly and disabled people can use automated mobility scooters from their homes, via public transportation, and in commercial facilities, such as shopping malls and hospitals. Therefore, the applicable areas for this study are assumed to be pedestrian areas, such that shopping malls and hospitals are pedestrian areas not restricted to specific travel lanes. These areas are expected to contain multiple pedestrians. In addition, as aforementioned, a mobility scooter is considered a pedestrian; hence, it moves in the same way as other pedestrians in pedestrian areas. Therefore, it is critical to cluster and track objects using sensors in crowded scenarios.

Furthermore, when using 2D-LiDAR for autonomous mobility, handling dynamic objects becomes more difficult and challenging compared to 3D-LiDAR, particularly for challenging scenarios such as multiple objects and crowds [4, 5]. Therefore, object tracking under 2D-LiDAR must involve the estimated values of the state to perform data optimization. In the framework of state estimation-based tracking, reliable parameter estimation methods include particle filtering (PF) and Kalman filtering (KF) [6]. In strong nonlinear systems, the accuracy and stability of the PF algorithm are advantageous [7]. However, as the number of particles in the PF increases, the computing cost increases exponentially, and the actual performance is also affected severely. Moreover, because the KF performs sufficiently when the mathematical model of the system is relatively simple and the estimated values follow a Gaussian distribution, this study employs the KF as an efficient mathematical state estimation and model for dynamic objects (e.g., pedestrians).

This study proposes an optimized clustering method based on a 2D-LiDAR low-cost scheme to improve the clustering and tracking accuracy in multi-pedestrian scenarios such as hospitals and shopping malls. The proposed method is applicable to motion scenarios, excluding sudden state changes in objects and extremely crowded scenarios. By improving the clustering accuracy, problems such as object coupling in other sensors (e.g., cameras) can be addressed via sensor fusion. Furthermore, this approach can positively impact later object classification, point cloud learning, etc.

2. Related Work

Owing to its long-range, wide-angle, and high-quality data, 3D-LiDAR has become crucial in most autonomous mobility platforms; however, it is difficult to implement on general/small platforms owing to its expensive equipment and high computational cost. Hence, 2D-LiDAR was employed as a low-cost solution. In actual application scenarios, when the ground is uneven, the single use of 2D-LiDAR cannot satisfy the actual application demand. However, considering that vehicles are mostly driven on flat roads, although 2D-LiDAR alone can basically meet the demand, the sparse point cloud makes the data unstable (sometimes vacant), which makes it difficult and challenging to extract useful information [8, 9].

To address the problem of sparse point clouds, Saito et al. [10] adopted the interpolation method using the sensor fusion of a camera and LiDAR based on the search range of depth data, and RGB camera data were identified for corresponding points. However, this method requires additional cameras and high-precision calibration beforehand and is limited to 3D-LiDAR only. Shan et al. [11] adopted deep learning in point clouds to increase the resolution and improve the sparse problem at far distances. However, this approach is slow and difficult to apply in real time and relies on learning large-scale scenes at high computational costs.

For point cloud processing, an effective clustering of obstacle information obtained by LiDAR can classify disordered point clouds from intuitive obstacles (shape, location, etc.) and classes for better environment perception [12]. For 2D-LiDAR, the accuracy and precision of clustering are crucial with the advantages of a low computational cost and sparse point cloud features.

Chen et al. [13] used 2D-LiDAR to extract pedestrian features and SVM for tracking. The sparse point clouds are clustered for feature extraction and pedestrians are finally tracked using KFs. However, a constant threshold is adopted in the initial clustering, and inaccurate clustering results directly influence subsequent tracking and classification. Sun et al. [14] proposed an improved Euclidean clustering method [15] and set different thresholds based on distance to improve clustering accuracy. However, it is difficult to obtain the optimal clustering threshold at each moment because the point cloud density varies in the time domain owing to object motion, even in the same distance range.

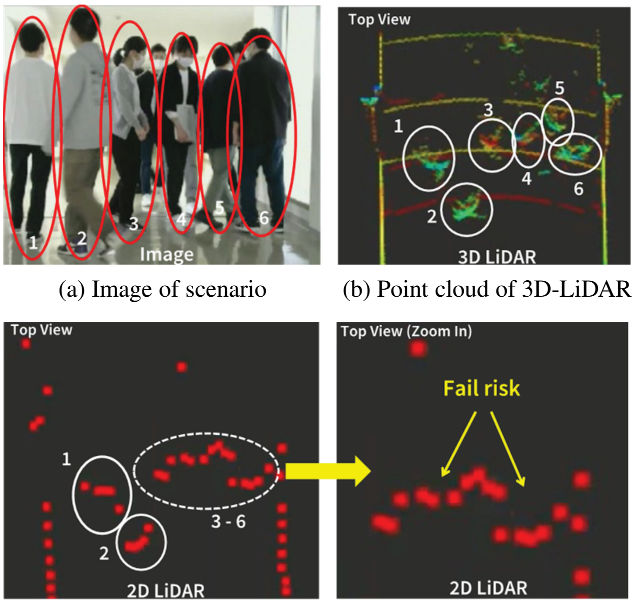
In addition, Kaleci et al. [16] proposed 2DLaserNet in clustering-based and regular deep learning, to improve the accuracy of the clustering and classification of 2D-LiDAR point clouds. However, the high computational cost of training point clouds and the highly efficient improvement in accuracy are negligible.

3. Issues and Purposed

Clustering is an important preprocessing task for object classification (recognition) and tracking. Improvements based on clustering are only uncertain [14, 15], and clustering results directly affect tracking accuracy [13]. Hence, we focused our optimal method on consistency with the tracking and post-processing of clustering. In addition, for a low computational cost [16], sparse point cloud processing in multiple pedestrian areas without training requires highly accurate clustering and tracking. The features of threshold variation in the time domain owing to dynamic objects must also be considered.

In previous studies, 3D-LiDAR exhibited satisfactory accuracy for typical multiple pedestrian scenarios (1–3 people). However, 2D-LiDAR has a higher risk of clustering failure owing to its low point-cloud density and accuracy. Therefore, it is necessary to develop a method that does not depend on the number of point clouds. **Fig. 1** presents 2D and 3D point clouds for a typical crowd scenario in actual applications. As illustrated in **Fig. 1(b)**, 3D-LiDAR provides rich point-cloud data, where the point clouds corresponding to the six objects in **Fig. 1(a)** are clearly distinguishable. However, for the same scenario, the performance of 2D-LiDAR in a sparse point cloud is illustrated in **Fig. 1(c)**. Among the same six objects, objects 1 and 2 can be accurately identified; however, for objects 3–6, the boundary between them is unclear. The risk of clustering failure in a crowd is at the moment when the objects are close to each other, similar to the case in objects 3–6. This is because the optimal value of the clustering threshold differs depending on the presence of multiple pedestrians. Hence, the scenario with multiple pedestrians is challenging for 2D-LiDAR and also crucial.

As aforementioned, although 2D-LiDAR can reduce the cost of adoption as an alternative, the amount of information is limited and the failure of clustering can severely affect the effectiveness of subsequent processing,



(a) Image of scenario (b) Point cloud of 3D-LiDAR

(c) Point cloud of 2D-LiDAR (d) Fail risk of 2D-LiDAR

Fig. 1. 2D- vs. 3D-LiDAR performance for each scenario.

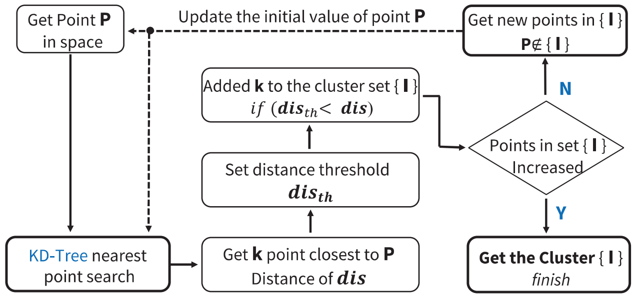


Fig. 2. Structure of Euclidean clustering.

i.e., tracking. In addition, clustering failure can lead to tracking failure owing to the inability to handle changing scenarios caused by dynamic changes, such as multiple pedestrians. Therefore, an improved method and solution for these scenarios are required to ensure optimal performance in the 2D-LiDAR.

4. Proposed Method

To address the issue of low tracking accuracy due to clustering failure, this study proposes KF and a variable threshold-based framework for 2D-LiDAR, to perform simultaneous optimal clustering and tracking of pedestrians in a simple motion model (modeling simple, no state abrupt change). By considering the consistency of clustering and the KF, clustering and tracking were optimized simultaneously. Then, the clustering threshold (tolerance) was varied according to the features of the point cloud in each scenario.

While obtaining optimal tracking results using the corresponding clustering thresholds, the clustering and tracking accuracies can be simultaneously improved.

4.1. Theoretical and Methodology

4.1.1. Euclidean Clustering

Among the several prior studies on clustering methods for point clouds, the Euclidean clustering, k-means [17], and DBSCAN [18], are widely used. In addition, the attribute-based Euclidean clustering is adapted in this study, considering that the actual requirement of the point cloud and critical information is mainly distance [19].

Euclidean clustering is based on Euclidean distance as the judgment criterion and has advantages including a simple structure, fast speed, and optimal performance. The principle is that for a point $\boldsymbol{P}$ in space, $\boldsymbol{k}$ points nearest to point $\boldsymbol{P}$ are searched in the domain via the KD-Tree [18], among which those with a distance $\boldsymbol{dis}$ less than the set threshold $\boldsymbol{dis}_{th}$ will enter set $\{\boldsymbol{I}\}$ ($\boldsymbol{dis}_{th} < \boldsymbol{dis}$; points within a certain distance threshold are grouped into a set). These procedures are iterated N times until non-new points within the current threshold are selected in set $\{\boldsymbol{I}\}$. Accordingly, its structure is shown in **Fig. 2**.

The two points required to calculate the Euclidean distance include the current point $\boldsymbol{n}$ and search point $\boldsymbol{n}+\mathbf{1}$, which correspond to the 2D point clouds $\boldsymbol{P}(x,y)$ and $\boldsymbol{k}(x,y)$ in **Fig. 2**, respectively. The two points entering the clustering set $\{\boldsymbol{I}\}$ are denoted as $\boldsymbol{I}_n$, $\boldsymbol{I}_{n+1}$, and the corresponding Euclidean distance is expressed as $\boldsymbol{dis}(\boldsymbol{I}_n, \boldsymbol{I}_{n+1})$. The Euclidean distance between the current $\boldsymbol{n}(x_n, y_n)$ and search $\boldsymbol{n}+\mathbf{1}(x_{n+1}, y_{\boldsymbol{n}+\mathbf{1}})$ can then be calculated as Eq. (1).

$$\boldsymbol{dis}\,(\boldsymbol{I}_n, \boldsymbol{I}_{n+1}) = \sqrt{(x_{n+1}-x_n)^2 + (y_{n+1}-y_n)^2}. \quad . \quad (1)$$

Moreover, the main issue with Euclidean clustering is the setting of a fixed distance threshold [20]. When scenarios such as distance or object motion change, the point cloud density changes accordingly. If a suitable threshold cannot be selected, fixed thresholds will lead to clustering failure, which severely affects the final tracking results.

4.1.2. KF and Modeling

Here, we use KF as the basis for tracking, and the tracked clusters represent added labels. As an optimal linear filter, KF is widely used in classical fields such as route planning, object tracking, and navigation owing to its advantages of efficiency, stability, and real-time performance [21]. They are also employed in new fields such as image detection, recognition, and sensor fusion [22]. The KF application must satisfy three basic assumptions: 1) the applicable dynamic system must be linear, 2) the observations are linear combinations of states, and 3) the noise satisfies a normal distribution (Gaussian distribution).

Moreover, we assume that the motion of dynamic objects is normal and constant; in addition, there is no sharp acceleration or deceleration. Therefore, this KF model is limited to pedestrians with sharp acceleration and deacceleration, such as stopping or changing direction. Subsequently, dynamic model processing and estimation are

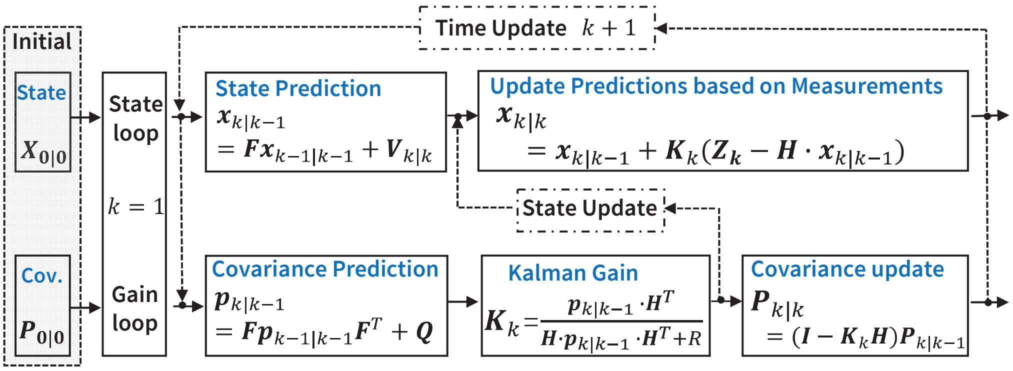


Fig. 3. State and gain update in KF ALOG.

performed for each point cloud frame as a unit and the time is extremely short; hence, the state model adopted here is approximately linear and the process noise satisfies the normal distribution. Similarly, the measurement noise of LiDAR satisfies a normal distribution for a constant-speed model in a very short time.

KF adopts a linear system state equation to estimate the state of the system from its inputs and outputs and is essentially a recursive algorithm, each of which comprises two main parts: estimation and measurement update [21]. When calculating the loop and outputs, a distinction can be made between the state- and gain-calculation loops. The main calculation equation involved in the KF is contained in the loop algorithm presented in **Fig. 3**.

In state prediction, the prediction model of the system state $\boldsymbol{x}_k$ (prior $\boldsymbol{x}_{k|k-1}$ and posterior $\boldsymbol{x}_{k|k}$ predicted values) and the measurement equation $\boldsymbol{Z}_k$ can be expressed as in Eqs. (2) and (3).

$$\boldsymbol{x}_{k|k-1} = \boldsymbol{F} \cdot \boldsymbol{x}_{k-1|k-1} + \boldsymbol{V}_{k|k}, \quad \dots\dots\dots (2)$$

$$\boldsymbol{Z}_k = \boldsymbol{H}_k \boldsymbol{x}_k + \boldsymbol{W}_k, \quad \dots\dots\dots (3)$$

where $\boldsymbol{F}$ denotes the process state transfer matrix, similar to Eq. (4), and the 2D-LiDAR refresh frequency of Δt is the same as 0.2 s ($\Delta \boldsymbol{t} = 0.2$ s). At this point, the state vector of the dynamic objects (pedestrian) can be expressed as $\boldsymbol{x}(k) = [\boldsymbol{x}_k \; \boldsymbol{y}_k \; \dot{\boldsymbol{x}}_k \; \dot{\boldsymbol{y}}_k]^T$, and the state of the system can be expressed in matrix form as Eq. (5):

$$\boldsymbol{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \dots\dots\dots (4)$$

$$\begin{bmatrix} \boldsymbol{x}_{k|k-1} \\ \boldsymbol{y}_{k|k-1} \\ \dot{\boldsymbol{x}}_{k|k-1} \\ \dot{\boldsymbol{y}}_{k|k-1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{x}_{k-1|k-1} \\ \boldsymbol{y}_{k-1|k-1} \\ \dot{\boldsymbol{x}}_{k-1|k-1} \\ \dot{\boldsymbol{y}}_{k-1|k-1} \end{bmatrix} + \begin{bmatrix} \boldsymbol{V}_x \\ \boldsymbol{V}_y \\ \boldsymbol{V}_{\dot{x}} \\ \boldsymbol{V}_{\dot{y}} \end{bmatrix}. \quad (5)$$

The process model of dynamic objects is set at a constant velocity, and $\boldsymbol{V}(\boldsymbol{V}_x, \boldsymbol{V}_y)$ represents the process noise (state noise), based on a normal distribution with mean 0 and covariance matrix $\boldsymbol{Q}(\boldsymbol{Q}_x, \boldsymbol{Q}_y)$ in the x-y direction, denoted as $\boldsymbol{V} \sim (0, \boldsymbol{Q})$. In addition, the acceleration noise factor $\boldsymbol{q}$ in the constant-velocity process was defined as $\boldsymbol{q} = 0.0001$. Matrix $\boldsymbol{Q}$ can be expressed as Eq. (6).

$$\boldsymbol{Q} = \begin{bmatrix} Q_x & 0 & 0 & 0 \\ 0 & Q_y & 0 & 0 \\ 0 & 0 & Q_{\dot{x}} & 0 \\ 0 & 0 & 0 & Q_{\dot{y}} \end{bmatrix} = q \times \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 & 0 & 0 \\ 0 & \frac{1}{2}\Delta t^2 & 0 & 0 \\ 0 & 0 & \Delta t & 0 \\ 0 & 0 & 0 & \Delta t \end{bmatrix}. \quad \dots\dots\dots (6)$$

In the measurement equation $\boldsymbol{Z}_k$, because 2D-LiDAR obtains the pedestrian's position information directly, there is no transformation relationship with the observed value. The velocity is an indirect value calculated based on position. The change matrix is denoted as $\boldsymbol{H}$, and is expressed as Eq. (7).

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad \dots\dots\dots (7)$$

Similar to the system model, the measurement model also contains noise and is denoted by $\boldsymbol{W}$, which is normally distributed with a mean of 0 and covariance matrix in the x-y direction with $\boldsymbol{R}(\boldsymbol{R}_x, \boldsymbol{R}_y)$ of the measurement noise as $\boldsymbol{W} \sim (0, \boldsymbol{R})$.

In addition, r denotes the accuracy of the measurement and the accuracy of 2D-LiDAR is $r = 0.02$ (≤ 10 m, 2%). The matrix $\boldsymbol{W}$ can be expressed as Eq. (8).

$$\boldsymbol{W} = \begin{bmatrix} R_x & 0 & 0 & 0 \\ 0 & R_y & 0 & 0 \\ 0 & 0 & R_{\dot{x}} & 0 \\ 0 & 0 & 0 & R_{\dot{y}} \end{bmatrix} = \begin{bmatrix} r^2 & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & \frac{r^2}{\Delta t} & 0 \\ 0 & 0 & 0 & \frac{r^2}{\Delta t} \end{bmatrix} \dots (8)$$

4.1.3. Initialization of Model

The initialization of the state and covariance matrix in the KF is also a key point, which directly affects the convergence speed and estimation accuracy. The initial relative distance of the target object by LiDAR is adopted as the initial state value $x_{0|0}$ of the KF, as in Eq. (9), where $(x_{0|clu}, y_{0|clu})$ denotes the initial state value from clustering (clu) and $(\dot{x}_{0|mov}, \dot{y}_{0|mov})$ is the average adult moving (mov) speed of 1.35 m/s.

$$x_{0|0} = \begin{bmatrix} x_{0|clu} \\ y_{0|clu} \\ \dot{x}_{0|mov} \\ \dot{y}_{0|mov} \end{bmatrix} \quad \text{. . . . . (9)}$$

The covariance of the model is P, and the initialization $P_{0|0}$ is the distance uncertainty of clustering. According to the average width of the average human body (0.43 m), the uncertainty in the x-y direction (P_{x_0}, P_{y_0}) is expressed with an error radius of 0.215, and $P_{0|0}$ is defined by Eq. (10).

$$P_{0|0} = \begin{bmatrix} P_{x_0} \\ P_{y_0} \\ P_{\dot{x}_0}\left(\dfrac{P_{x_0}}{\Delta t}\right) \\ P_{\dot{y}_0}\left(\dfrac{P_{y_0}}{\Delta t}\right) \end{bmatrix} \quad \text{. . . . . (10)}$$

4.1.4. Evaluation of Clustering Parameter

To evaluate the clustering results and accuracy, the point clouds were compared based on the difference between the measured and predicted results. While obtaining the difference value, a normal distribution was obtained for the point cloud at (x_g, y_g). Then, to reduce the position uncertainty, the optimal threshold was selected based on the occupancy grid map (OGM), which is often used to distinguish whether the detection area is occupied by objects for matching and comparing [23]. Here, the difference values were calculated as the difference in the OGM between the measured and predicted results. In addition, for the 2D normal distribution $f(x_g, y_g)$, if the random variables x_g, y_g are independent, the correlation function is 0, as expressed in Eq. (11).

$$f(x_g, y_g) = \frac{1}{2\pi\sigma_{x_g}\sigma_{y_g}} e^{-\frac{1}{2}\left\{\frac{(x_g-\mu_{x_g})^2}{\sigma_{x_g}^2} + \frac{(y_g-\mu_{y_g})^2}{\sigma_{y_g}^2}\right\}} \quad \text{. . (11)}$$

In addition, the sum of absolute difference (SAD) of the difference of each OGM was calculated at the same threshold as the standard for evaluating the optimal clustering threshold.

4.2. Approach

The specific approach flow is presented in **Fig. 4**, which primarily contains clustering, prediction, and evaluation modules and performs tracking under the selection of optimal tolerance values to achieve optimized clustering and tracking simultaneously.

Moreover, the density features of the point cloud differ from each other owing to factors such as object class and distance; hence, setting the tolerance of clustering

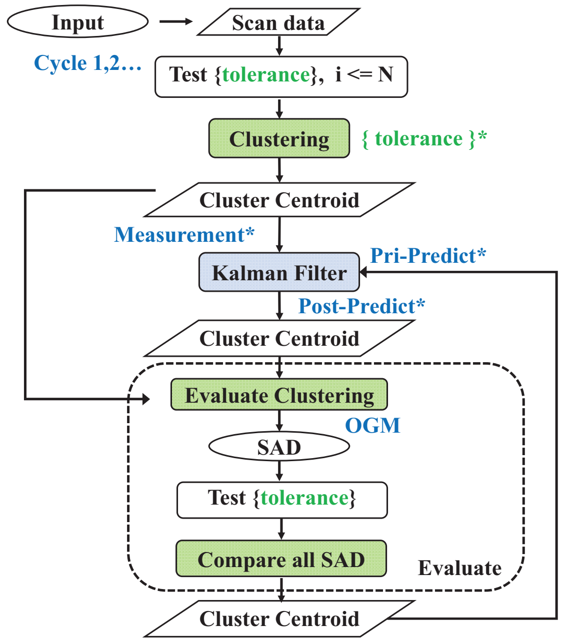


Fig. 4. Workflow of approach and processing.

for each scene is critical. For objects considered multiple pedestrians [24, 25], setting the tolerance excessively small may ascertain one object as multiple clusters, and the contrary may identify multiple objects as one cluster. Therefore, clustering tolerance was set as variable parameter. Regarding the variable tolerance, we cluster several candidate tolerance values to obtain the optimal based on the KF tracking. Here, the tracking points are the centroids of the clusters, which are also used as measurements (predicted KF values). The SAD value was adopted as the evaluation parameter by comparing the difference between the final predicted and measured values. The values of several candidate tolerances corresponding to the smallest SAD were selected as the optimal threshold. The evaluation process based on SAD in each variable tolerance is illustrated in **Fig. 5**.

The dataflow and specifics of each step in the proposed method are presented in **Algorithm 1**.

- Applying the scenario in actual pedestrian environments with no complex movements (rapid changes).
- The loop block in the algorithm performs N loops in the parameter selection of the clustering part, where the candidate tolerance and $\{$/tolerance, $N\}$ changes each time. $N = 10 = \{0.15, 0.20, 0.25, 0.30, 0.40, 0.45, 0.50, 0.55, 0.60\}$.
- The limitation of the approach is that the optimal tolerance in the scenario is in the candidate tolerance.
- The OGM is calculated based on the probability of the point cloud position distribution.
- Results under the minimum value of SAD (similarity of $\{$/centroid$\}$; $\{$/estimated_centroid$\}$) are selected as the final optimal prediction and clustering.

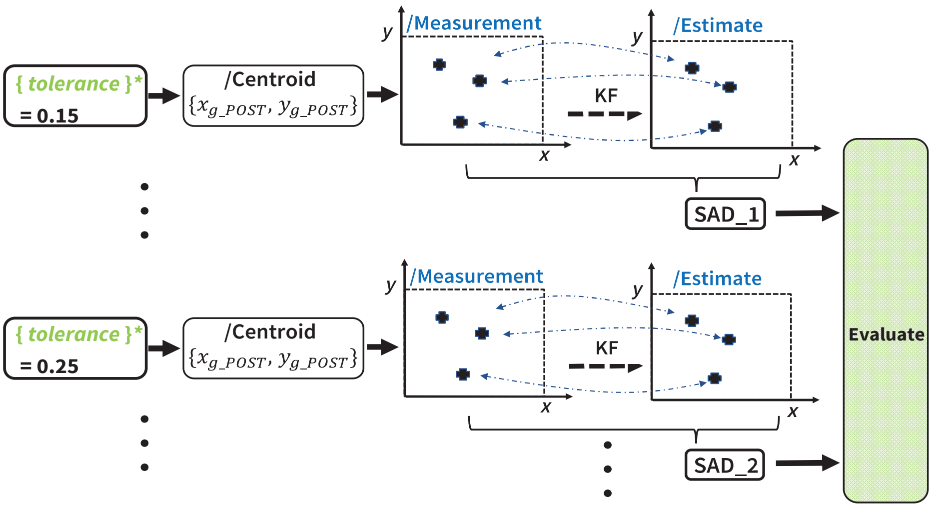


Fig. 5. Clustering evaluation module.

Algorithm 1. KF and clustering in 2D Pts

For cycle $n \in N$:

Initialize input *as* $n = 1$

Init-input *Point cloud obtained by 2D-LiDAR*

$(x_{L_1}, y_{L_1}), (x_{L_i}, y_{L_i}), i \in \{/Points\_Scan\}$

- **Euclidean clustering**

$(x_{g_1}, y_{g_1}), (x_{g_j}, y_{g_j}), j \in \{/Centroid\}$

- **Measurement input** *as* $\mathbf{Z}_n$, $n = 1$

$(x_{m_1}, y_{m_1}), (x_{m_j}, y_{m_j}), j \in \{/Centroid\}$

- **Kalman filtering**

$(x_{g\text{-}Po}, y_{g\text{-}Po}, P_{g\text{-}Po}) \in \{/Estimate\_Centroid\}$

input *as* $n = 2, 3, \ldots, N$

- **Measurement input** *as* $\mathbf{Z}_n$, $n = 2, 3, \ldots, N$
- **Kalman filtering**

$(x_{g\text{-}Po}, y_{g\text{-}Po}, P_{g\text{-}Po}) \in \{/Estimate\_Centroid\}$

Alternate estimate

$(x_{Al\_POST}, P_{Al\_POST})$

Evaluate

OGM, $(x_{g\text{-}Po}, y_{g\text{-}Po}), (x_{g_j}, y_{g_j})$

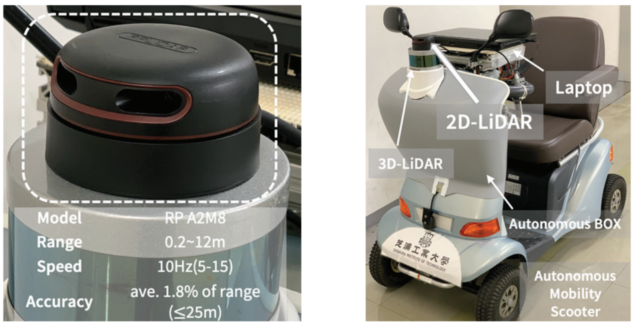
Compare

SAD, $(x_{POST}, y_{POST}) \in \{/Fitness\_Score\_Vec\}$

Output

(x_{POST}, y_{POST})

End For



(a) RP 2D-LiDAR (b) Data acquisition platform

Fig. 6. Equipment and platform.

5. Experiment

The mobile platform for the development of an automated mobility scooter, which was also used in this experiment, and its external sensing part are described. In the sensing part, the effectiveness of the optimization method in the proposed scheme to replace 3D-LiDAR using 2D-LiDAR with a lower import fee and computational cost was verified and analyzed.

Moreover, the properties of the pedestrian objects and limited scenarios were defined and analyzed based on a simple motion model. Furthermore, the results of the optimized clustering and effectiveness of the filtering model were verified. Then, the proposed variable threshold was analyzed and compared with the clustering results using the conventional fixed-threshold method.

5.1. Equipment Description

During the experiments, Euclidean clustering algorithms based on the C++ version processed the 2D-LiDAR and ran on a laptop computer with Intel Core i5 and 16 GB memory on a robot operating system (ROS) [26] in Linux.

Regarding the specifications of sensing parts, the RP LiDAR A2, as illustrated in **Fig. 6(a)**, has a single channel, time of flight (TOF) ranging principle, measurement range of 0.2–12 m, range accuracy of average 1.8% ($\leq$ 25 m), 360° horizontal, and default rotation rate of 10 Hz (5–15 Hz). It is assumed that the angular and linear velocities of the LiDAR are smooth and continuous over time, without abrupt changes.

The autonomous mobility scooter was adapted for the data acquisition platform, and the placement of sensors and other devices is illustrated in **Fig. 6(b)**.

5.2. Scenarios and Assumption

In describing and defining the motion characteristics and scenarios of moving objects (e.g., pedestrians), there are generally five main elements: the number of pedestrians, movement pattern, movement characteristic, relative speed, and movement trajectory in an actual environment.

1) Number of pedestrians: two types of scenarios with single and multiple cases.
2) Movement patterns categorized into five types of movement scenarios: straight ahead, separated, approaching, crossing, and irregular.
3) Movement characteristics categorized into three types of moving scenarios: constant speed, sudden stop, and variable speed [27].
4) Relative velocity classified into two types of scenarios, depending on whether movement exists on the observation side (LiDAR side) or not.
5) Movement trajectory in the real environment classified into two scenarios: where the data is in the real environment and where the data is in the direction of the simulation or the data acquirer.

In selecting and setting up the scenario for the experiments, we employed a typical scenario commonly occurring during the field of testing (FOT) at large shopping malls (no instructions directed to pedestrians) [2]. In this scenario, there are multiple pedestrians in the forward direction (the same direction as the mobility scooter), and the pedestrian movement can be approximated as straight ahead. In addition, the velocity of pedestrians can be defined as a constant velocity for a short time span. Furthermore, because the velocity of the mobility scoter is close to that of the pedestrian (the pedestrian is faster), the relative velocity and displacement of the pedestrian are small while the mobility scoter remains in motion. This makes it difficult to understand the results of the analysis and the validation of the proposed motion method.

Therefore, to verify the effectiveness of the model under the condition of motion speed where the displacement is easier to understand, a one-sided motion with the fixed mobility scooter was adopted. Accordingly, the scenario selected for this experiment was as follows:

1) Number of pedestrians: multiple (three pedestrians).
2) Movement pattern: straight ahead.
3) Movement characteristics: constant speed (in a short time span) owing to the processing in each point cloud frame, which is extremely short [28].
4) Relative speed: no movement on the observation side (LiDAR side).
5) Movement trajectory: real environment.

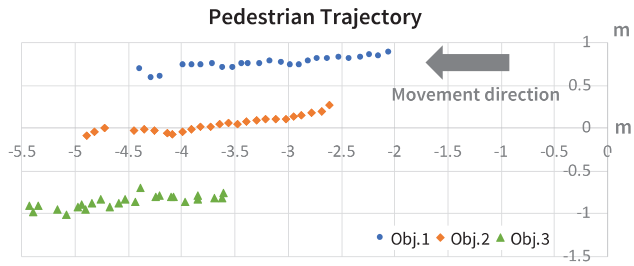


Fig. 7. Pedestrian movement trajectory by centroid of objects.

Figure 7 presents the trajectory of pedestrians (objects 1–3) in the scenario using the centroid of each object. Therefore, the actual point cloud data for clustering are distributed around each centroid, and it can be determined that the pedestrians tend to approach each other in all frames when the width of the pedestrians (including the luggage in their hands) is considered. In addition, pedestrians tend to approach each other, and the distance between them changes over time.

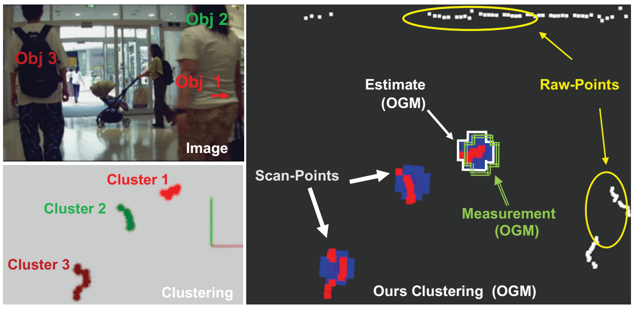
5.3. Results

The results for the actual pedestrian motion scene are presented in **Fig. 8**, where the object reference image from the RGB camera is illustrated in the upper left corner of each figure, the results of clustering are shown in the lower part, and the right presents the results of obtaining the optimal threshold by the OGM, which contains the OGM results for the prediction model and measurements, and the optimal threshold (tolerance = 0.2, 0.25, and 0.35) at each moment is obtained.

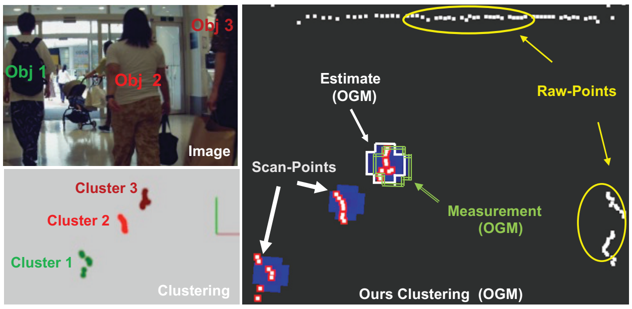
Figure 8 presents the results for a tolerance values of 0.2 (**Fig. 8(a)**), 0.25 (**Fig. 8(b)**), and 0.35 (**Fig. 8(c)**). The scan and raw points are the point cloud data obtained from the 2D-LiDAR and original point cloud, respectively. Accordingly, a general selection filter is used to select the forward area point cloud for data analysis and prediction according to the scenario requirements.

The results of the OGM exhibit no significant deviation in the distribution of the final estimated values (dotted lines) and measurement values (double solid line). This verifies that the method combining 2D-LiDAR and KF can correctly predict the central distribution after clustering and obtain the optimal threshold, which validates the effectiveness of the model.

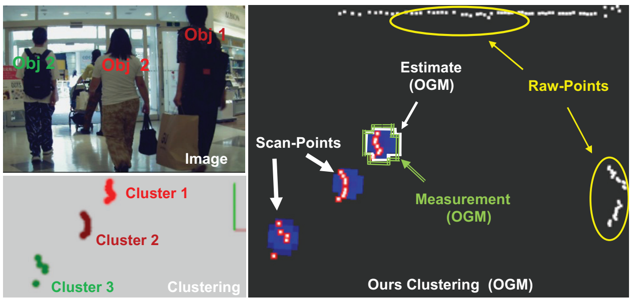
Furthermore, the Kalman gain (K_k) was evaluated to determine the effectiveness of the model structured in the KF, as illustrated in **Fig. 9**. The proposed method evaluates the difference between the observed and predicted KF values, assuming that KF modeling has been performed successfully. Therefore, K_k was utilized to evaluate the KF model, and the results demonstrated that the variation in K_k decreased rapidly as the number of steps increased. This confirms that the confidence weights were quickly transferred to the estimation. Finally, K_k reached 0.07, which increased the prediction accuracy of the proposed KF model and verified the effectiveness of the model.



(a) Tolerance = 0.2



(b) Tolerance = 0.25



(c) Tolerance = 0.35

Fig. 8. Results of probability distribution at optimal clustering thresholds (tolerance) by OGM.

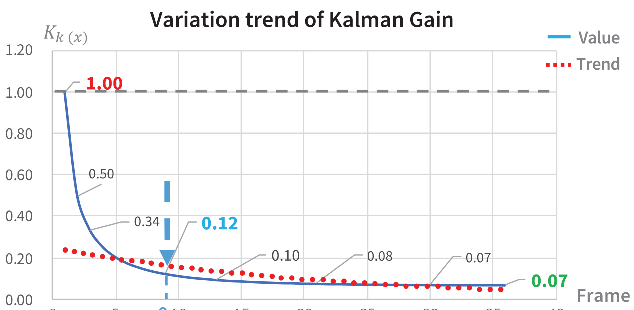


Fig. 9. Evaluation of Kalman modeling using Kalman gain.

Moreover, for the variable tolerance parameter in the proposed method, the specific values are presented in **Fig. 10**, from which it can be observed that the tolerance is adjusted for different scenarios. In the experiment, the corresponding optimal tolerance took a maximum value of 0.35, a minimum value of 0.15 (a deviation of 0.2), and frequent changes to shift within a short period of 40 frames. Therefore, it can be verified that the use of fixed tolerances often leads to optimal clustering results for each frame and increases the risk of clustering failure. Hence, employing variable tolerances is critical and beneficial for optimal clustering.

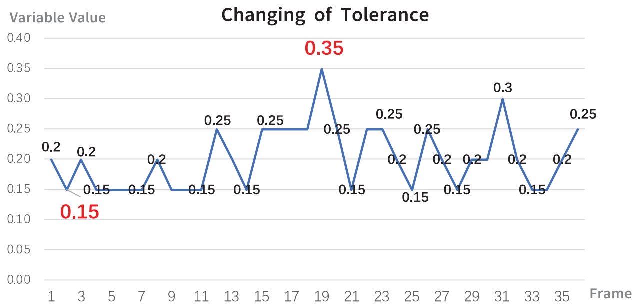


Fig. 10. Shift of variable tolerance.

Table 1. Evaluation of clustering accuracy by SSE.

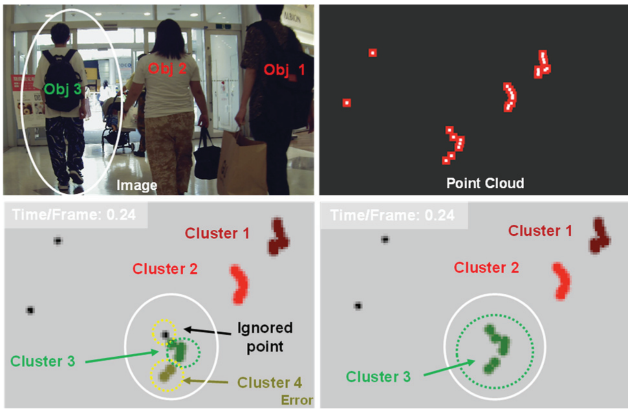
Method	w	w/o				
Tolerance	Variable	0.15	0.2	0.25	0.3	0.35
Sum of SSE	**113.6**	–	115.2	127.2	129.4	129.5

5.4. Analysis

The proposed method was adopted to evaluate the selection of the optimal clustering threshold (tolerance). In the clustering of point clusters, the smaller the variation of point clusters within the same cluster (object), the better clustering is considered. Therefore, the sum of squared errors (SSE), which is the sum of the squares of the distances between each point in the same cluster and the centroids of that cluster, was employed as an evaluation parameter. The smaller the SSE value, the smaller the variation within a cluster, and the more accurate the clustering results [29].

In the evaluation, the SSE was calculated for each cluster within each frame, and the evaluation was based on the sum of the total frames (36 frames). Therefore, the evaluation value is the "sum of the SSEs," and the number of clusters in each frame is the same. The obtained results are presented in **Table 1**, which demonstrates that the sum of the SSEs obtained by the proposed method is the smallest (= 113.6) compared with the conventional method with a fixed clustering threshold (tolerance = 0.20, 0.25, 0.30, 0.35). In addition, when the tolerance was 0.15, evaluation by the sum of SSEs was not performed because the clustering failed. Hence, the effectiveness and accuracy of our method can be verified and the tolerance for optimal clustering can be selected.

As mentioned in **Table 1**, clustering fails when a fixed threshold of 0.15 is used. As illustrated in **Fig. 11(a)**, the failed clustering result indicates that the point cloud of the three subjects is classified into four clusters and one point outside the clusters (ignored point). In the same situation (same frame), while using our proposed variable



(a) Tolerance = 0.15 (without ours (fixed))

(b) Tolerance = 0.25 (with ours (variable))

Fig. 11. Clustering results compare with ours vs. w/o ours in the same situation.

clustering threshold algorithm, the optimal threshold was selected as 0.25, as illustrated in **Fig. 11(b)**. Object 3 was clustered into one cluster (cluster 3), indicating that the correct result was obtained. Therefore, our method was verified to be superior and more effective than the conventional method.

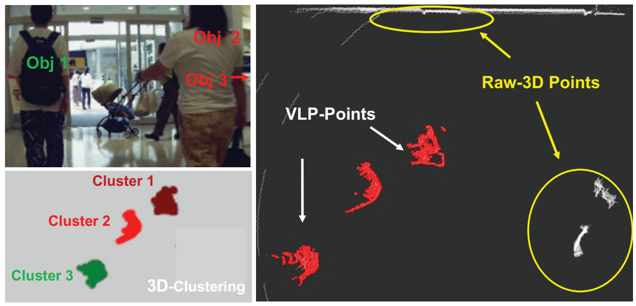
6. Discussion

The core of the method proposed in this study is the optimization of the clustering threshold. In the analysis, it is verified that setting a threshold (tolerance) to variable parameters results in a higher clustering accuracy than the fixed threshold type of Euclidean clustering and also effectively avoids the risk of clustering failure due to improper threshold selection. Moreover, the applicability of our proposed method framework is not only limited to Euclidean clustering but is also valid for other clustering methods with variable thresholds.

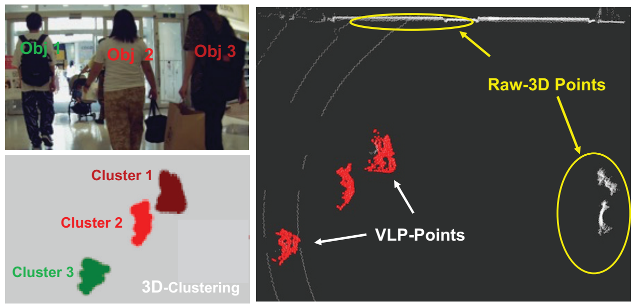
Regarding the applicable data type of the proposed method, the 2D-LiDAR point cloud was adopted as the object in this study, and the same method can be applied to 3D-LiDAR and verified. The clustering results of the 3D-LiDAR data were collected at the same time as those of the 2D data, as illustrated in **Fig. 12**.

Figure 12(a) presents the results for a tolerance of 0.2 and **Fig. 12(b)** of 0.25 while applying the 3D-LiDAR data. The results of 3D-clustering (clusters 1–3) prove that our method can be applied to 3D-LiDAR and obtain the correct clustering results based on the variable tolerance scheme. Moreover, tolerance was adjusted while obtaining the optimal clustering results for application in 3D-LiDAR. Then, the corresponding optimal threshold was verified between maximum and minimum values of 0.25 and 0.15 (a deviation of 0.1), respectively, as illustrated in **Fig. 12(c)**.

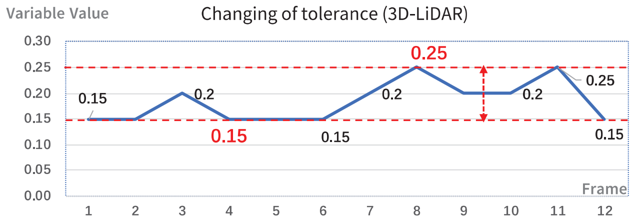
In addition, the tolerance values for optimal clustering were selected from a predefined candidate set, as described in **Algorithm 1**, and the effect was verified by experimental results. However, the method is limited in that the optimal threshold cannot be selected if it does not exist in the candidate parameter set. Hence, it is necessary to change the clustering tolerance value from a candidate set to a random set structure.



(a) Tolerance = 0.2



(b) Tolerance = 0.25



(c) Shift of variable tolerance

Fig. 12. Results of proposed method applicable to 3D-LiDAR.

Furthermore, the effectiveness of the proposed method was verified in multiple pedestrians (three pedestrians) in a shopping mall for the experiment. However, the density characteristics of a crowd differ for each scenario and location. Hence, when the threshold features can be set and differentiated according to the scenario or location as the base, the proposed method should be widely applicable.

7. Conclusion

Here, a low-cost alternative method for optimal clustering applicable to 2D-LiDAR for mobility was proposed. Optimal clustering and tracking were performed simultaneously by combining variable clustering tolerances and KF. In addition, OGM was employed to optimize the clustering tolerance and improve the performance and accuracy of clustering.

The experimental data were collected in an actual shopping mall center environment. For normal motion scenarios with multiple pedestrians in front of the line direction, the proposed method was verified to obtain more accurate clustering and tracking results than the conventional fixed tolerance method. Moreover, the evaluation of the Kalman gain demonstrated the effectiveness of the Kalman filter model. The evaluation of the final clustering accuracy using SSE verified that our method exhibits a higher clustering accuracy than the conventional method.

Furthermore, in the future, the KF modeling should be extended to the extended Kalman filter (EKF), unscented Kalman filter (UKF), and PF for complex scenarios where the object undergoes rapid acceleration (deceleration) or directional changes. Furthermore, to improve the clustering results and expand the range of applicability, the selection of the optimal tolerance from a random set according to a normal distribution should be considered.

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References:

[1] M. Z. Azmi and T. Ito, "Artificial potential field with discrete map transformation for feasible indoor path planning," Applied Sciences, Vol.10, No.24, pp. 1-13, 2020.

[2] C. Yamamoto et al., "Development and Usability Evaluation of Automated Set Box for Mobility Scooter," 2022 JSAE Annual Congress Proc., pp. 111-124, 2022.

[3] Y. Tazaki and Y. Yokokohji, "Outdoor autonomous navigation utilizing proximity points of 3D Pointcloud," J. Robot. Mechatron., Vol.32, No.6, pp. 1183-1192, 2020.

[4] R. Ren et al., "Large-scale outdoor SLAM based on 2D Lidar," Electronics, Vol.8, No.6, Article No.613, 2019.

[5] M. G. Ocando et al., "Autonomous 2D SLAM and 3D mapping of an environment using a single 2D LIDAR and ROS," 2017 Latin American Robotics Symp. and Brazilian Symp. on Robotics, pp. 1-6, 2017.

[6] G. Welch et al., "An introduction to the kalman filter," Proc. of SIGGRAPH, Course, Vol.8, No.41, pp. 27599-23175, 2001.

[7] P. K. Gaddigoudar et al., "Pedestrian detection and tracking using particle filtering," Int. Conf. on Computing, Communication and Automation (ICCCA), pp. 110-115, 2017.

[8] S. Gidel et al., "Pedestrian detection method using a multilayer laserscanner: application in urban environment," 2008 IEEE/RSJ Int. Conf. on Intelligent Robots and Systems, pp. 173-178, 2008.

[9] T. Taipalus et al., "Human detection and tracking with knee-high mobile 2D LIDAR," Int. Conf. on Robotics and Biomimetics, pp. 1672-1677, 2011.

[10] M. Saito et al., "Interpolation method for sparse point cloud at Long-range using LiDAR and camera sensor fusion," Trans. of Society of Automotive Engineers of Japan, Vol.53, No.3, pp. 598-604, 2022.

[11] T. Shan et al., "Simulation-based lidar super-resolution for ground vehicles," Robotics and Autonomous Systems, Vol.134, Article No.103647, 2020.

[12] L. Zheng et al., "The obstacle detection method of UAV based on 2D Lidar," IEEE Access, Vol.7, pp. 163437-163448, 2019.

[13] J. Chen et al., "Pedestrian detection and tracking based on 2D Lidar," 2019 6th Int. Conf. on Systems and Informatics (ICSAI), pp. 421-426, 2019.

[14] Z. Sun et al.,"An improved lidar data segmentation algorithm based on Euclidean clustering," Proc. of the 11th Int. Conf. Modeling, Identification and Control (ICMIC), pp. 1119-1130, 2020.

[15] N. Bouhmala, "How good is the Euclidean distance metric for the clustering problem," 2016 5th IIAI Int. Congress on Advanced Applied Informatics (IIAI-AAI), pp. 312-315, 2016.

[16] B. Kaleci et al., "2DLaserNet: A deep learning architecture on 2D laser scans for semantic classification of mobile robot locations," Engineering Science and Technology, an Int. J., Vol.28, Article No.101027, 2022.

[17] G. D. Tipaldi et al., "Motion clustering and estimation with conditional random fields," Proc. of the 2009 IEEE/RSJ Int. Conf. on Intelligent Robots and Systems, pp. 872-877, 2009.

[18] A. Asvadi et al., "DepthCN: Vehicle detection using 3D-LIDAR and ConvNet," Proc. IEEE 20th Int. Conf. on Intelligent Transportation Systems (ITSC), pp. 1-6, 2017.

[19] S. Bandyopadhyay et al., "A point symmetry-based clustering technique for automatic evolution of clusters," IEEE Trans. on Knowledge and Data Engineering, Vol.20, No.11, pp. 1441-1457, 2008.

[20] E. Güngör et al., "Distance and density based clustering algorithm using Gaussian kernel," Expert Systems with Applications, Vol.69, pp. 10-20, 2017.

[21] O. Li et al., "Kalman filter and its application," 2015 8th Int. Conf. on Intelligent Networks and Intelligent Systems (ICINIS), pp. 74-77, 2015.

[22] S. Shen et al., "Feature Detection Based on Virtual Gradient Using Sensor Fusion for Low-Resolution 3D LiDAR," 2021 10th IEEE CPMT Symp. Japan (ICSJ), pp. 41-44, 2021.

[23] M. Schütz et al., "Occupancy grid map-based extended object tracking," 2014 IEEE Intelligent Vehicles Symp., pp. 1205-1210, 2014.

[24] A. F. Foka et al., "Predictive autonomous robot navigation," IEEE/RSJ Int. Conf. on Intelligent Robots and Systems, Vol.1, pp. 490-495, 2002.

[25] X. Hu et al., "Road centerline extraction in complex urban scenes from LiDAR data based on multiple features," IEEE Trans. on Geoscience and Remote Sensing, Vol.52, No.11, pp. 7448-7456, 2014.

[26] M. Quigley et al., "ROS: An open-source robot operating system," ICRA Workshop on Open Source Software, Vol.3, No.2, pp. 1-6, 2009.

[27] W. G. Weng et al., "Cellular automaton simulation of pedestrian counter flow with different walk velocities," Physical Review E, Vol.74, No.3, Article No.036102, 2006.

[28] Y. Luo et al., "PORCA: Modeling and planning for autonomous driving among many pedestrians," IEEE Robotics and Automation Letters, Vol.3, No.4, pp. 3418-3425, 2018.

[29] J. Qi et al., "K-means: An effective and efficient K-means clustering algorithm," 2016 IEEE Int. Conf. on Big Data and Cloud Computing (BDCloud), Social Computing and Networking (SocialCom), Sustainable Computing and Communications (SustainCom), pp. 242-249, 2016.

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- M. Saito et al., "Infrastructure Alert System Based on Intersection Monitoring Using Measurement Sensor and Infrared Camera," Trans. of Society of Automotive Engineers of Japan, Vol.53, No.6, pp. 1253-1259, 2022.
- M. Saito, S. Shen, and T. Ito, "Interpolation method for sparse point cloud at long distance using sensor fusion with LiDAR and camera," 2021 IEEE CPMT Symp. Japan (ICSJ), pp. 116-117, 2021.

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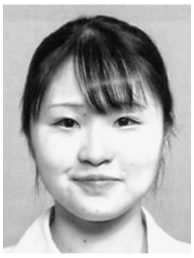

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- M. Z. Azmi and T. Ito, "Artificial potential field with discrete map transformation for feasible indoor path planning," Applied Sciences, Vol.10, No.24, pp. 1-13, 2020.
- T. Ito and S. Sudo, "Raising driver attentiveness with unconscious learning effect," Procedia Computer Science, Vol.126, pp. 891-898, 2018.

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